

Unit : 1 : Foundation Agile & Devops

Modern software

- Modern software development required fast delivery better quality & continuous improvement.
- Traditional software development approaches often suffer from long development cycle, poor communication, delayed feedback & higher failure rates. to address these challenges to major approaches, emerge.

1. Agile:- It focuses on iterative software development & customer collaboration.

2. DevOps:- It focuses on collaboration between development and operation teams to automate and accelerate software delivery.

What is Agile?

Agile is software development methodology that include continuous delivery of software, customer collaboration, adaptation of changing requirements and incremental iterative development.

instead of building a system at once agile develops software in small increments for sprints.

Agile manifesto :-

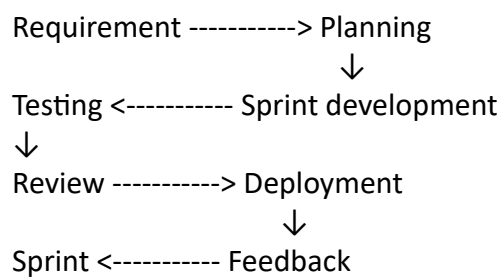
1. Individual and interactions over Process and tools.
2. working software over comprehensive documentary.
3. customer collaboration over contract negotiation.
4. Responding to change over following the Plan.

*** Agile Principles :-**

1. Customer satisfaction through early & continuous delivery.
2. Welcomes changing requirements.

3. Deliver working software frequently.
4. Business People & developers work together daily.
5. Build Projects around motivated individuals.
6. Face to Face Communication is most effective.
7. Working Software is the Primary measure of Progress.
- 8.
9. Continuous attention to the technical excellence.
10. Simplicity is essential.
11. Self-organizing team creates the best Solutions.
12. Teams regularly reflect & improves.

Agile lifecycle :-



Lean Thinking :-

Lean thinking originated from Together Production system (TPS)
It's goal is to deliver maximum Customer value with minimum waste.
Lean focuses on improving efficiency & eliminating activities that don't add values.

Lean Principles :-

- > Identify values & understand what Customers need.
- > map the value stream by identifying all steps required.
- > Create a flow which ensure throughout smooth execution.
- > Establish a bridge by producing only when needed.
- > Peruse perfection by continuously improving the process.

DevOps :-

- What is Devops ?
 - Devops is a combination of development + operations, cultural set of Practices and tools that enables orgination to delivery software rapidly and reliably.

The three ways of Devops

(1) The flow :- It focuses on development, testing, deployment & then Customer

-> The main objective is to reduce the error, automate the work flows & faster delivery

Ex :- CI/CD Pipeline automatically deploys software

(2) Feedback :- It's creates rapid feedback tools which contains operations then development & then improvement.

-> It's benefits by fixing the bug faster and improve reliability.

Ex :- monitoring tools instantly report failures

(3) Continue learning :- It encourages experimentation, innovation and continuous learning.

Ex :- Conducting trials after every sprint.